

$$\Omega = (x_1, x_2) = \{(0,0), (0,1), (1,0), (1,1)\}$$

$$2.2.1 \quad p(x_1, x_2) = \left(\frac{2}{3}\right)^{x_1+x_2} \left(\frac{1}{3}\right)^{2-x_1-x_2}$$

$$\begin{aligned} P(x_1) &= \sum_{x_2 \in \Omega} \left(\frac{2}{3}\right)^{x_1+x_2} \left(\frac{1}{3}\right)^{2-x_1-x_2} = \left(\frac{2}{3}\right)^{x_1+0} \left(\frac{1}{3}\right)^{2-x_1-0} + \left(\frac{2}{3}\right)^{x_1+1} \left(\frac{1}{3}\right)^{2-x_1-1} \\ &= \left(\frac{2}{3}\right)^{x_1} \left(\frac{1}{3}\right)^{2-x_1} + \left(\frac{2}{3}\right)^{x_1+1} \left(\frac{1}{3}\right)^{1-x_1} \\ &= \left(\frac{2}{3}\right)^{x_1} \left(\frac{1}{3}\right)^{1-x_1} \left[\left(\frac{1}{3}\right)^1 + \frac{2}{3} \right] = \left(\frac{2}{3}\right)^{x_1} \left(\frac{1}{3}\right)^{1-x_1} \end{aligned}$$

$$\begin{aligned} P(x_2) &= \sum_{x_1 \in \Omega} p(x_1, x_2) = \left(\frac{2}{3}\right)^{x_1+x_2} \left(\frac{1}{3}\right)^{2-x_1-x_2} + \left(\frac{2}{3}\right)^{1+x_2} \left(\frac{1}{3}\right)^{1-x_2} \\ &= \left(\frac{2}{3}\right)^{x_2} \left(\frac{1}{3}\right)^{1-x_2} \left[\left(\frac{1}{3}\right)^1 + \left(\frac{2}{3}\right)^1 \right] \\ &= \left(\frac{2}{3}\right)^{x_2} \left(\frac{1}{3}\right)^{1-x_2} \end{aligned}$$

$$y_1 = x_1 - x_2$$

$$y_2 = x_1 + x_2$$

$$\Rightarrow \begin{cases} \frac{y_1+y_2}{2} = x_1 = x_1 = \frac{y_1+y_2}{2} \\ \frac{y_1+y_2}{2} - y_1 = x_2 = x_2 = \frac{y_2-y_1}{2} \end{cases}$$

$$P(y_1, y_2) = P_{x_1, x_2} \left(\frac{y_1+y_2}{2}, \frac{y_2-y_1}{2} \right) |J| = \left(\frac{2}{3}\right)^{\frac{y_1+y_2}{2}} \left(\frac{1}{3}\right)^{2-\frac{y_1+y_2}{2}} \left(\frac{1}{2}\right)$$

$$2.2.2 \quad P(x_1, x_2) = \frac{x_1 x_2}{46} \begin{cases} x_1 = 1, 2, 3 \\ x_2 = 1, 2, 3 \end{cases} \quad \begin{matrix} y_1 = x_1 y_2 \\ y_2 = x_2 \end{matrix} \quad \begin{matrix} x_1 = \frac{y_1}{y_2} \\ x_2 = \frac{y_2}{y_2} \end{matrix}$$

$$P(y_1, y_2) = \begin{cases} \frac{1}{36} & y_1 \in \{1, 2, 3, 4, 6, 9\} \\ 0 & \text{if else} \end{cases} \quad y_1 = x_1, y_2 = x_1 x_2$$

$$P_{x_1}(y_1) = \sum_{y_2} \frac{y_1}{36} = \frac{y_1}{36} = P_{y_1} = \begin{cases} \frac{1}{36} & y_1 = 1 \\ 0 & \text{else} \end{cases}$$

$$2.2.3 \quad (X_1, X_2) \sim h(x_1, x_2) = \begin{cases} 2e^{-x_1-x_2} & 0 < x_1 < x_2 < \infty, \\ 0 & \text{elsewhere.} \end{cases}$$

$$Y_1 = 2X_1, \quad Y_2 = X_2 - X_1$$

$$X_1 = \frac{Y_1}{2}, \quad X_2 = Y_2 + \frac{Y_1}{2}$$

$$|J| = \begin{vmatrix} 1/2 & 0 \\ 1/2 & 1 \end{vmatrix} = 1/2$$

$$f_{Y_1, Y_2}(y_1, y_2) = \begin{cases} e^{-y_2-y_1} & \begin{cases} 0 < y_1 < \infty, 0 < y_2 < \infty \\ 0 < y_1 < \infty \end{cases} \\ 0 & \text{elsewhere} \end{cases}$$

$$0 < \frac{y_1}{2} < \frac{2y_2+y_1}{2} < \infty$$

$$0 < y_1 < 2y_2 + y_1 < \infty \quad y_1 - 2y_2 < y_1$$

$$-\frac{y_1}{2} < \left(\frac{2y_2+y_1}{2} \right)$$

$$-2y_1 - 2y_2 = -y_2 - y_1$$

$$2.2.4 \quad (X_1, X_2) \sim h(x_1, x_2) = \begin{cases} 8x_1x_2 & 0 < x_1 < x_2 < 1 \\ 0 & \text{otherwise.} \end{cases}$$

$$Y_1 = X_1/X_2, \quad Y_2 = X_2$$

$$X_1 = Y_1Y_2, \quad X_2 = Y_2$$

$$f_{Y_1, Y_2}(y_1, y_2) = \begin{cases} 8(y_1y_2)(y_2) & 0 < y_1 < 1 \\ 0 & \text{if else} \end{cases}$$

$$|J| = \begin{vmatrix} y_2 & y_1 \\ 0 & 1 \end{vmatrix} = y_2$$

$$0 < y_1y_2 < y_2 < 1$$

$$y_1 < \frac{1}{y_2} < 1$$

$$y_1y_2 < y_2 < 1$$

$$f_{Y_1, Y_2}(y_1, y_2) = \begin{cases} 8y_1y_2^3 & 0 < y_2 < 1 \\ 0 & \text{if else} \end{cases}$$

$$\begin{cases} x_1 = y_1 - y_2 \\ y_1 = x_1 + x_2 \end{cases} \quad x_2 = x_2 \quad |J| = \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} = 1$$

2.2.5 $f_{x_1, x_2}(x_1, x_2) = \begin{cases} f_{x_1, x_2} & -\infty < x_1 < \infty \\ 0 & \text{elsewhere} \end{cases}$

(a) $f_{y_1, y_2} = \begin{cases} f_{x_1, x_2}(y_1 - y_2, y_2) & -\infty < y_1 - y_2 < \infty, -\infty < y_2 < \infty \\ 0 & \text{elsewhere} \end{cases}$

(b) $f_{y_1}(y_1) = \int_{-\infty}^{\infty} f_{x_1, x_2}(y_1 - y_2, y_2) dy_2$. To show this

Recall that the marginal distribution of $f_{y_1}(y_1)$ can be computed as $f_{y_1}(y_1) = \int_{(-\infty, \infty)} f_{x_1, x_2}(y_1, y_2) dy_2$

from (a) we derived that under the transformation $f_{x_1, x_2}(y_1, y_2) = f_{x_1, x_2}(y_1 - y_2, y_2)$, thus we have

$$f_{y_1}(y_1) = \int_{(-\infty, \infty)} f_{x_1, x_2}(y_1 - y_2, y_2) dy_2. //$$

2.2.6 x_1 and $x_2 \sim f_{x_1, x_2}(x_1, x_2) = \begin{cases} \exp\{- (x_1 + x_2)\} & 0 < x_1 < \infty \\ 0 & \text{elsewhere} \end{cases} \quad z=1, 2$

(a) $f_{y_1}(y_1) = \int_{(-\infty, \infty)} \exp\{-(y_1 - y_2 + y_2)\} dy_2$

$$= \int_{-\infty}^{\infty} \exp\{-y_1\} dy_2 = \exp\{-y_1\} \int_{-\infty}^{\infty} 1 dy_2$$

$e^{-y_1} \sim \Gamma(1, 1)$

(b) $\int_0^{\infty} e^{-y_1 t} y_1^0 dt = \frac{1}{1+t} = \frac{1}{1+t^2}, t < 1$

$$X_1 - Y = X_1, Y \quad Y = Y_2 X_2$$

2.2.5

$$Y_1 = X_1 + X_2$$

$$(X_1, X_2) \sim \begin{cases} f_{X_1, X_2}(x_1, x_2) = 2e^{-(x_1 + x_2)}, & 0 < x_1 < x_2 < \infty \\ 0 & \text{elsewhere} \end{cases}$$

$$f_{Y_1}(y_1) = \int_{y_1/2}^{y_1} f_{X_1, X_2}(y_1 - y, y) dy$$

$$0 < y_1 - y < y < \infty \\ 0 < y_1 < 2y < \infty$$

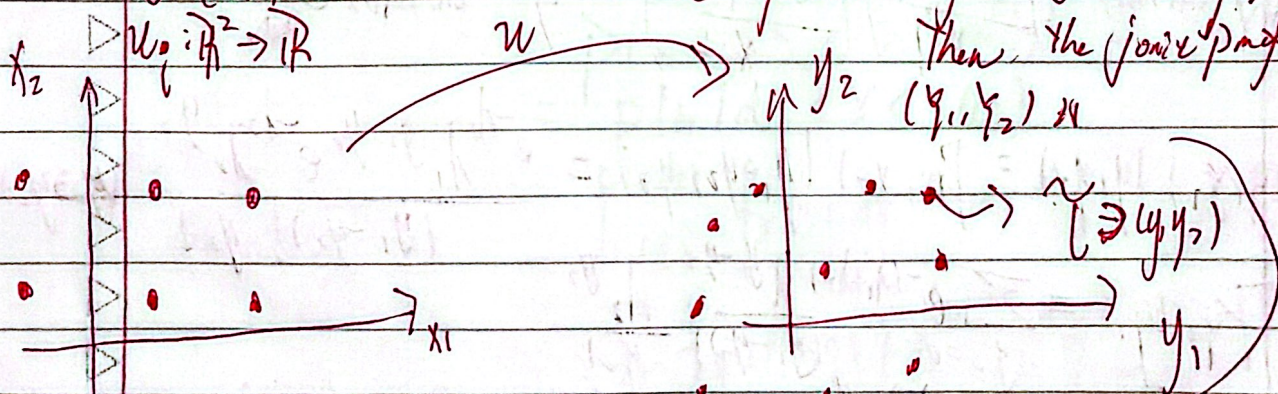
$$= \int_{y_1/2}^{y_1} 2 \exp\{-(y_1 - y + y)\} dy$$

$$= \int_{y_1/2}^{y_1} 2 \exp\{-y_1\} dy = 2y_1 e^{-y_1} \Big|_{y_1/2}^{y_1}$$

$$= 2y_1 e^{-y_1} - 2\left(\frac{y_1}{2}\right) e^{-y_1}$$

$$= 2y_1 e^{-y_1} - y_1 e^{-y_1} = y_1 e^{-y_1} \quad y_1 > 0$$

Week 2 Let (X_1, X_2) be discrete r.v.'s with joint pmf $p_{X_1, X_2}(x_1, x_2)$ and support $\mathcal{S}$. Let $Y_1 = u_1(X_1, X_2)$ and $Y_2 = u_2(X_1, X_2)$ where $u = (u_1, u_2)$ is a one to one function from $\mathcal{S}$ to $\mathcal{T}$. Then, the joint pmf of (Y_1, Y_2) is



the set $(X_1, X_2) \in \mathcal{S}$

$$P_{(Y_1, Y_2)}(y_1, y_2) = P_{(X_1, X_2)}(u^{-1}(y_1, y_2))$$

Homework 2. 3.6.9

Let f have an F -distribution with parameters r_1, r_2 .
Argue that $1/f$ has an F -distribution with parameters r_2 and r_1 .

$$f \sim F(r_1, r_2), \quad f_F(r_1, r_2; x)$$

$$f_F = \begin{cases} \frac{\Gamma(\frac{r_1+r_2}{2}) (\frac{r_1}{r_2})^{r_1/2} (x^{r_1/2} - 1)}{\Gamma(\frac{r_1}{2}) \Gamma(\frac{r_2}{2}) (1 + \frac{r_1 x}{r_2})^{(\frac{r_1+r_2}{2})}} & 0 < x < \infty \\ 0 & \text{elsewhere.} \end{cases}$$

Let $Q = \frac{1}{f}$ so $f_Q(q) = f_F(\frac{1}{q}) \frac{1}{q^2}$

Also since $f = \frac{U/r_1}{V/r_2} \sim F$ when $U, V \sim \chi^2$

then $Q = \frac{1}{f} = \frac{V/r_2}{U/r_1} \sim F(r_2, r_1)$ since $V, U \sim \chi^2$

3.6.10 $F(r_1=5, r_2=10; x)$ $P(X \leq a) = 0.05, P(X \leq b) = 0.9$
 $P(a < X < b) = 0.90$
 $a = 0.21119, b = 3.386$

3.6.11 $T = \frac{W}{\sqrt{V/r}}$ $W \sim N(\mu=0, \sigma^2=1)$ Show $T^2 \sim F(r_1=1, r_2=r)$.
 $V \sim \chi^2(df=r)$.

$W \sim Z \sim N(0, \sigma^2=1)$ We know that $Z^2 = \chi^2(df=1)$

Thus $T = \frac{Z^2/1}{V/r} = \frac{\chi^2/1}{\chi^2/r} = F(r_1=1, r_2=r)$

Q.E.D.